Forgetting and completeness: the centre of the passage from Theorem 3.11 to Corollary 3.12 in inter-universal Teichmüller theory

Toshihisa Urahigashi
123026.urahigashi.toshihisa@gmail.com

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Independent researcher. This work was carried out personally and is unrelated to any institutional affiliation.

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Abstract

We formulate the passage from Theorem 3.11 to Corollary 3.12 in inter-universal Teichmüller theory (IUT) as a problem of *completeness* in the sense of abstract interpretation. A forgetful map $\alpha : X \to A$ is complete for a query $Q : X \to Y$ if Q factors through α, that is, if the abstract data determine the answer to Q. The same forgetting can be complete for one query and incomplete for another: sign abstraction is complete for the sign of a product but not for the sign of a sum, and interval abstraction drops the correlation between two uses of the same datum. Completeness is therefore a relation between an abstraction and a query, and local completeness at each interface does not in general compose.

We first state, in a form checked in Lean, the criterion for completeness, its restoration by a tracked invariant, and the sufficiency and—on extremal data with no slack—the sharpness of scale preservation. We then read the multiradial machinery as one diagram: a tower of forgettings from initial Θ-data through fundamental groups, mono-theta environments, prime-strips, unit and value-group parts and realified Frobenioids down to log-volume, together with the reverse mono-analytic reconstruction flow and the horizontal $\mathfrak{log}$-link and Θ-link. In the cited theory T that fixes the log-volume content cited in the proof of Corollary 3.12, the Corollary 3.12-shaped conclusion does not follow from T alone, and it does follow once the scale defect $\Delta = (s-1)\,\mathrm{vol}(\theta)$ satisfies $\Delta \leq 0$; preservation itself is the vanishing $\Delta = 0$, that is, $s = 1$ when $\mathrm{vol}(\theta) \neq 0$. On extremal data with no slack the conclusion is equivalent to $\Delta \leq 0$. Relative to this route, therefore, a sufficient condition for the ABC-type conclusion reduces to the non-trivial completeness problem of whether the real scale is preserved through the whole transport, which contains forgetful steps. This is not an unconditional equivalence with the ABC conjecture itself.

An audit of the relevant interfaces in IUT I–IV and in [FrdI, EtTh, AbsTopIII] finds each of them locally closed: Tate parameter reconstruction, cyclotomic synchronization, the discrete valuation step, unit-side indeterminacy, within-copy volume normalization, and the three-tier classification of degrees under realification. What remains is a single square: whether the actual construction supplies a lift of the Θ-link, given between bare prime-strips, to marked prime-strips preserving generators and scale. The reduction also shrinks the formalization target: what has to be formalized is not all of IUT, but only the interfaces needed to decide the sign of Δ. This paper decides neither Corollary 3.12 nor the ABC conjecture.

Keywords: inter-universal Teichmüller theory, Corollary 3.12, abstract interpretation, completeness, forgetful functor, dependency problem, realification, scale, Lean.

Marks used below: [FRAME] framing or translation; [EST] proved here or verified in the source; [OPEN] unresolved; [H] the present author's reading or interpretation.

1 Introduction: two questions

The dispute over Mochizuki's inter-universal Teichmüller theory is often discussed as a single question. At least the following two must be separated.

Question S (specification).
Does the conclusion of Corollary 3.12 follow from the explicitly stated log-volume content of the results cited in its proof alone?

Question C (construction).
Does the actual construction of IUT preserve, from end to end, the correlations (initial data, labels, choices of roots, real scale) that are not recorded in the summaries of the cited properties?

[EST] The preceding study [U26] showed, by countermodels checked in Lean, that the answer to Question S is no for the theory obtained by fixing the cited range, and that a weak form of the conclusion does follow. The present paper does not repeat that result. Its purpose is to state precisely, in the language of forgetting and completeness, where the gap lives and what would have to be extracted to make Question C decidable.

[FRAME] The central reading of this paper is the following. The multiradial machinery consists of a sequence of forgetful steps, each discarding information stage by stage, together with a reverse flow of mono-analytic reconstructions that recover discarded structure. At each interface one can ask for which queries the answer is not lost. The query of Corollary 3.12 is not a local query at a single interface; it is a correlation query that runs across the same initial data and the shared markings. Hence each interface may be locally exact while the composite is incomplete for that query.

[H] This reading is consistent with Mochizuki's own exposition. The source repeatedly states that distinct arithmetic holomorphic structures are not to be confused, that they are related only through the shared shadow of prime-strips, and that the cyclotome and the value-group part are shielded from unit-side indeterminacy. In the language of this paper, this is the claim that the construction is designed to track the labels. The point at issue is whether that claim is closed as a proof at the level of the cited statements.

In this paper the remaining point is called the *centre* (Japanese: *honmaru*, the innermost keep of a castle), not a wall. Seen from the specification side it appears as a wall that the cited properties alone cannot cross. Once it has been isolated as a completeness problem, it appears instead as the central proposition that remains to be proved; if that completeness holds, this route reaches the ABC conjecture. "Centre" is thus the name of the single point where the remaining work is concentrated, with no connotation of unattainability.

2 Simple examples first: when it is preserved and when it breaks

2.1 Forgetting signs: preserved for products, broken for sums

Consider the forgetful map $\mathrm{sign} : \mathbb{Z} \to \{-1, 0, 1\}$ that keeps only the sign of an integer, and the abstraction $\alpha(a, b) = (\mathrm{sign}\, a, \mathrm{sign}\, b)$ of a pair of integers. For one and the same α, completeness is opposite for the two queries:

Query	Result	Reason / witness
Is $\operatorname{sign}(ab)$ determined?	complete	$\operatorname{sign}(ab) = \operatorname{sign}(a)\operatorname{sign}(b)$
Is $\operatorname{sign}(a+b)$ determined?	incomplete	$(1,-1)$ and $(2,-1)$ share the image $(+,-)$ but the signs of the sums are 0 and $+$

[EST] This contrast is checked in Lean as `completeFor_sign_mul` and `not_completeFor_sign_add`. The same sign contrast (complete for multiplication, incomplete for addition) is also the introductory example of [GRS00]. What this shows is that completeness is not a property of an abstraction alone but a relation between an abstraction and a query. A forgetting that keeps multiplicative structure may be complete for multiplicative queries and incomplete for queries involving addition. This is the smallest model for reading the steps of IUT in which addition is forgotten in the passage to prime-strips.

2.2 Interval arithmetic: forget the correlation and even $x - x$ is undetermined

Suppose a number x is known only to lie in the interval $[0, 1]$. Summarizing componentwise by intervals, the two uses (x, x) of the same x become $([0, 1], [0, 1])$, and the difference spreads to $[-1, 1]$. From the abstract image alone one cannot distinguish the pair $(0, 0)$ (difference 0) from the pair $(0, 1)$ (difference -1). This is the classical *dependency problem* [Moore] (for a modern textbook treatment see [MKC09]).

The cure is also classical. Attach a shared label to each occurrence—the noise symbols of affine arithmetic—so that the information "the two copies are the same datum" is carried along; then at least whether the difference is zero is decided again [SF97]. The point is that even if every local interval computation is exact, the end-to-end correlation query becomes incomplete at the moment the shared label is dropped.

Remark 2.1 (The pattern used repeatedly below). Local completeness fails to compose not because a computation is wrong, but because an interface summary does not record the correlation "the same thing was used twice". Restoration requires not a finer abstraction but carrying along an invariant that represents the lost correlation.

[H] In IUT, the counterparts of the shared label are the relativity to initial Θ-data, the LabCusp labels, the choices of roots and theta values, and finally the relative scale at realification. The question of this paper is whether the last of these, the scale, is preserved through an actual run.

3 The frame, stated rigorously

Abstract interpretation is the theory that organizes approximate computation by Galois insertions and abstract domains [CC77, CC79], and completeness has been studied there as the property that abstract computations accumulate no loss of information relative to concrete computations [GR97, GRS00]. Here we use neither the order structure nor fixpoints; we extract only the minimal factorization form that is needed. This is a deliberate simplification, and the statements of this paper are claimed only inside the simplified frame.

Definition 3.1 (Completeness)**.** For maps $\alpha : X \to A$ and $Q : X \to Y$, we say that α is *complete for* Q if there exists $\widehat{Q} : A \to Y$ with $Q = \widehat{Q} \circ \alpha$. In Lean this is `CompleteFor`.

Proposition 3.2 (Fibre criterion)**.** [EST] *If α is complete for Q, then $\alpha(x_1) = \alpha(x_2)$ implies $Q(x_1) = Q(x_2)$. Conversely, if Y is nonempty and Q is constant on every fibre of α, then α is*

complete for Q. Hence incompleteness is always witnessed by two states with the same abstract image and different answers.

Proposition 3.3 (Restoration by a tracked invariant)**.** [EST] *If Q factors through $x \mapsto (\alpha(x), c(x))$ and the invariant c is normalized to a constant c_0 on the states under consideration, then on those states $\alpha(x_1) = \alpha(x_2)$ implies $Q(x_1) = Q(x_2)$. In Lean,* `complete_on_normalized` *is proved without axioms.*

Proposition 3.4 (Reduction theorem: the scale defect)**.** [EST] *For the signed log-volume relation $\mathrm{vol}(q) = s \cdot \mathrm{vol}(\theta)$, put $\Delta = (s-1)\,\mathrm{vol}(\theta)$. Then:*

(i) *(Sufficiency) If $\mathrm{vol}(\theta) \le \mathrm{vol}(H)$ and $\Delta \le 0$, then $\mathrm{vol}(q) \le \mathrm{vol}(H)$.*

(ii) *(Equivalence with slack) $\mathrm{vol}(q) \le \mathrm{vol}(H)$ if and only if $\Delta \le \mathrm{vol}(H) - \mathrm{vol}(\theta)$. In particular, with no slack it is equivalent to $\Delta \le 0$.*

(iii) *(Preservation itself) If $\mathrm{vol}(\theta) \neq 0$, then $\Delta = 0$ if and only if $s = 1$.*

(iv) *(Sign translation) In the convention of [U26], where the relevant log-volumes are negative, $\Delta \le 0$ is equivalent to $s \ge 1$, and the countermodel lies on the side $s = 3/14 < 1$.*

In Lean these are `conclusion_of_nonpositive_scale_defect`*,* `conclusion_iff_defect_le_slack`*,* `scale_defect_eq_zero_iff_scale_eq_one` *and* `nonpositive_defect_iff_one_le_of_neg`*.*

The direction of a one-sided inequality depends on the parameterization (in the convention where a multiplier t acts on a nonnegative hull bound, the favourable side is $t \le 1$). We therefore take the convention-independent defect Δ as the main actor, and define preservation itself as $\Delta = 0$, i.e. $s = 1$.

Convention	Favourable side	Preservation	Countermodel
[U26]: $\mathrm{vol}(q) = s\,\mathrm{vol}(\theta)$, log-volumes negative	$s \ge 1$ ($\Delta \le 0$)	$s = 1$ ($\Delta = 0$)	$s = 3/14 < 1$ fails
Hull bound: $q_{\mathrm{vol}} \le t\,h_{\mathrm{vol}}$, nonnegative	$t \le 1$	$t = 1$	reciprocal parameterization

[FRAME] Propositions 3.3 and 3.4 are elementary; they are not deep theorems by themselves. Their role is to fix the logical shape of the claim: a failure of completeness is turned into the checkable question "which invariant is not being tracked?", and the additional principle needed for restoration can be named as a single item. The mathematical difficulty is thereby transferred to whether that additional principle is supplied by the actual construction.

[H] Reading an α off a text is itself not a theorem. What can be guaranteed here is the implication "complete for the stated α", together with the auditability of where each component of that α comes from in the source. Formalization does not remove this interpretive boundary; it concentrates it into one visible point.

4 Reading IUT as a diagram of forgettings

[FRAME] Read the multiradial machinery as the following tower of stages: from full arithmetic data, through tempered fundamental groups and mono-theta environments, prime-strips, unit parts

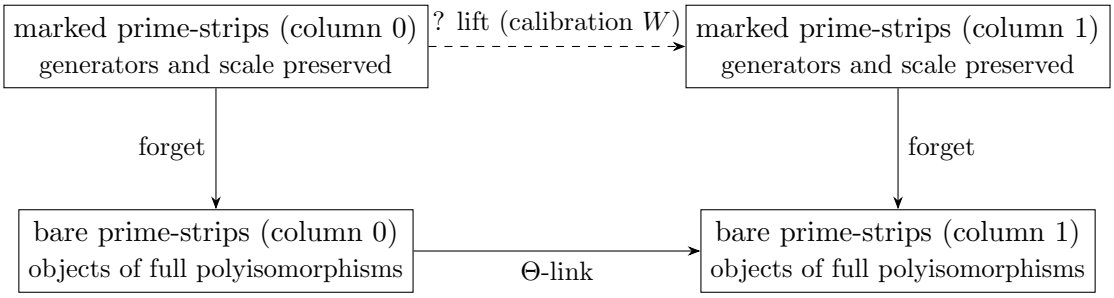


Figure 1: The missing square. The lower arrow exists as a full polyisomorphism in the sense of [IUT3, Def. 3.8(ii)]. The question is whether the actual construction supplies the upper arrow preserving the same initial-data calibration.

and value-group parts, and realified Frobenioids, down finally to the real number given by log-volume. Each stage forgets part of the structure. In the reverse direction, mono-analytic reconstruction recovers part of what was dropped. Horizontally there are the $\mathfrak{log}$-link inside a column, the Θ-link between columns, and the Kummer correspondences and evaluation maps.

That the lower arrow exists and that the upper arrow exists are different assertions. From the lower arrow alone there remains the relative scale c, the ratio of the readouts in the two columns. Transported once around a loop, c may be regarded as the holonomy of the loop; $c = 1$ is the condition that the cocycle is a coboundary, i.e. that there is a marking making every transport have scale 1. Since the scale of each copy is anchored absolutely, e.g. by $\mu_k(\mathcal{O}_k) = 1$, the number c is not a mere freedom of convention but an invariant of the chosen member.

[FRAME] In categorical language this is a lifting problem for a forgetful functor $U : \mathbf{Marked} \to \mathbf{Bare}$. A multiradial representation may be read as a functorial assignment from the log-theta-lattice to containers, and the question of Corollary 3.12 asks whether that assignment preserves one specific diagram, the comparison between columns. Category theory provides no magic proof here; it provides a language in which the squares to be checked can be listed finitely and the faces that are already closed can be separated from the one that is open.

5 Relative reduction to scale preservation: what is and is not equivalent

We separate the ranges in which the often-quoted phrase "an equivalent reformulation of a sufficient condition for ABC" is accurate. The claim made in this paper is the following relative reduction. In the cited theory T, which fixes the log-volume content cited in the proof of Corollary 3.12, and in the models formalized in that range:

(1) from T alone the strong, Corollary 3.12-shaped conclusion does not follow;

(2) if the scale-defect condition $\Delta \leq 0$ (preservation itself being $\Delta = 0$, i.e. $s = 1$) is added to T, the strong conclusion does follow (Proposition 3.4);

(3) on extremal data with no slack the conclusion is equivalent to $\Delta \leq 0$; hence for uniform derivations the scale condition is not only sufficient but sharp.

In this sense, relative to this route of IUT—from Theorem 3.11 through Corollary 3.12 to the numerical assembly of IUT IV, as assembled in the source—a sufficient condition for the ABC-type conclusion reduces to the completeness problem of whether the real scale is preserved through

the whole transport, which contains forgetful steps. That is, this scale-preservation problem is a sufficient condition for ABC, and IUT, along this route, reduces ABC to it. Is that preservation then solved, proved, refuted, or verified? The answer is no in every case (§§6–7).

What is *not* claimed is an unconditional equivalence between the ABC conjecture itself and this completeness problem. For an individual elliptic curve the height inequality may hold for reasons of slack or by a different proof route, in which case $\Delta \leq 0$ is not necessary. Equivalence holds only relative to the fixed cited theory and uniform derivations, or on extremal data.

[EST] Proposition 26 of [U26] contains the nucleus of this reduction: (a) the gap is not bridged inside the container alone; (b) identifying the two pilots yields the conclusion and $s = 1$; (c) from $s = 1$ and nonzero measure the conclusion follows, so that contrapositively a failure of the conclusion forces $s \neq 1$. The general factorization shape through the pair (α, s) is not a numbered proposition there, which is why it is presented separately here as Propositions 3.3 and 3.4.

[EST] In the sideways Lean formalization as well, the gap to the conclusion is bounded by the defect $\Delta = (\mathrm{linkScale} - 1) \cdot \mathrm{procVol}(\theta)$; the Corollary 3.12 shape is its $\Delta \leq 0$ point and preservation itself is the $\Delta = 0$ point. The same "defect $\leq$ slack" shape has been obtained in the repository of Project LANA, where Step (xi) is shown to be equivalent to adding the inequality $(s^{-1} - 1)a \leq V$ to a Theorem 3.11-level package (`xi_iff`) [lana]; this is the reciprocal parameterization of Proposition 3.4(ii). The equality (9-1) of [LANA, §9.2] states the same content from the equality side, as commutativity of a diagram. The contribution of the present paper is therefore not the discovery of this reduction itself, but its restatement in the language of completeness and the localization of what remains by an interface-by-interface audit. Within that audited decomposition, end-to-end completeness reduces to the liftability in the missing square. This is a reduction relative to the audited decomposition in which the other faces are closed; it is not an unconditional equivalence theorem about the whole construction.

6 The interface-by-interface audit

The following table summarizes the author's audit and formalization of the interfaces in IUT I–IV and [FrdI, EtTh, AbsTopIII], carried out against checked editions of the sources. "Closed" means that completeness for the local query at that interface was verified as a statement of the source or as a Lean theorem; it does not mean end-to-end completeness.

Interface	Kind	Status	Grounds
Identification of initial Θ-data	provenance	fixed by definition	A provenance condition on both theaters; not a condition on the choice of member [IUT3, Def. 3.8(ii)].
Curve to Tate parameter	reconstruction	closed	Intrinsic via j; Lean Tate reconstruction and the identification with $\log\|j(W)\|$.
Cyclotomic synchronization	reconstruction	closed	[EtTh, Cor. 2.19(i)]: finite-level Tate-twist identification; discrete ambiguity excluded.
Local monoid reconstruction	reconstruction	closed	[AbsTopIII, Prop. 3.2]: normalization by the unramified value-monoid generator; rigidity of the value monoid $\mathbb{N}$.
Unit-side indeterminacy	quotient	closed at readout	Ism acts by a unit scalar; the Galois part preserves at $e = 1$ and has only a displacement bound for $e > 1$.

Interface	Kind	Status	Grounds
Identification at realification	quotient	tiered	The three-tier $\mathbb{R}_{>0}/\mathbb{Q}_{>0}/1$ classification of [FrdI, Thm. 6.4]; [IUT2, Cor. 4.10(v)] states the lowest tier explicitly.
Evaluation isomorphism env $\cong$ gau	transport	closed within a theater	[IUT2, Cor. 4.5(v), Cor. 4.6(iv)(v)]: functorial in discrete data; not the cross-column side.
Value-part quotient and readout	quotient	closed	The radial readout factors through the quotient; element-level membership is out of scope.
Log-volume normalization	quotient	closed within a copy	$\mu_k(\mathcal{O}_k) = 1$, $\mu^{\log}(\mathfrak{m}^n) = -fn \log p$, agreeing with the cardinality normalization of [FrdI, Ex. 6.3].
Function-level theta values	transport	conditionally closed	When s, i, t lie in the base field they agree with the canonical value classes; a root-field version exists separately. Identification as functions remains.
Lift of the Θ-link	correspondence	**open**	No supplying statement for the calibration could be located in the audited range; the missing square of this paper.
Holomorphic hull and membership	interval hull	**open**	Identification of the source's holomorphic hull family with the same-run full-image family; only model-level results.
Numerical assembly of IUT IV	computation	closed given input	[IUT4, Thm. 1.10] assumes the situation of Corollary 3.12; the issue is on the input side.

[H] What this table shows is an accumulation of evidence that the audited interfaces are closed with one exception. Mathematically it does not guarantee that no unknown interface exists. The sentence "end-to-end completeness reduces to one square" must therefore be read relative to the decomposition and the audited range of this paper.

7 What would settle it

A document settling Question C positively must contain at least:

(1) an identifiable description of which member of the Θ-link is used in an actual run;

(2) a calibration, or an equivalent lattice-preserving lift, guaranteeing that this member computes both pilots from the same initial $q_{\underline{v}}$ in the two columns;

(3) an extraction, line by line, of where that guarantee comes from among the cited statements;

(4) for the holomorphic hull, an identification of the full-image family obtained in one run with the source's holomorphic hull family.

Conversely, a negative settlement at the construction level would require exhibiting a scale-changing transport that is realizable as a member of the actual construction, not an abstract countermodel

at the specification level. At present neither the positive extraction nor a construction-level counterexample has been presented in verifiable form.

The advantage: minimal formalization suffices

[FRAME] The practical advantage of this reduction is that the formalization target shrinks from "a Lean formalization of all of IUT" to "the minimal formalization needed to decide whether scale completeness holds". What is needed for the decision is only: (a) the log-volume content of the cited theory T; (b) the definition of the defect Δ and the reduction lemmas of Proposition 3.4; (c) at the missing square, the input-side definitions and the transport of generators for the actual Θ-link member between marked prime-strips; and (d) the statements of the interfaces that this square consumes directly. Every other interface can be black-boxed together with a certificate that it is complete for its query. If a statement cited as a black box itself hides the scale, this is of course insufficient, and minimality depends on the auditability of the interface statements. Even so, the work changes from re-reading everything to concentrating on the seams.

[FRAME] Categorical language makes the extraction target precise. Regarding the log-theta-lattice as a finitely presented diagram and a multiradial representation as a functor from that diagram to containers, the question of Corollary 3.12 becomes the commutativity of a specific square; as long as the other faces are closed, commutativity of the whole is equivalent to commutativity of the remaining face. As diagram chasing this is elementary, but expanding the definitions of the generating arrows returns one to the enormous construction. Here lies the double character of the problem: it looks provable, and at present it is non-trivial.

[H] This does not mean logical independence or Gödel-type unprovability. It is a concrete mathematical problem about a finite text and a finite diagram, decidable in principle by extraction and verification. It is open partly because it includes the interpretive question of whether the text pins down the construction uniquely enough for the question to be decided.

8 Position and non-claims

Relation to earlier observations

The nucleus "scale information is lost by the identifications" is not new. The report of Scholze–Stix [SS18] raised the identification issue in Corollary 3.12 at the specification level. Project LANA formulated the same point as the commutativity of the equality (9-1) [LANA, §9.2], and its Lean repository checks the missing inequality as `xi_iff` [lana]; for the mathematical nucleus of the reduction to a scale defect, priority of publication belongs to LANA. The contribution of the present paper and of the preceding formalization study [U26] is to cut that observation down to a machine-checkable incompleteness theorem with a fixed cited range, and then to rearrange it into interface-wise certificates and the missing square.

Relation to the author's earlier preprints

This paper is related to the following of the author's preprints. [U26] is the direct predecessor: the cited theory T and the shape of its Proposition 26 are taken from there, and the present paper restates that result in the language of completeness and connects it to the interface audit and the missing square. [UCS] is an essay reading IUT as "two computations of the same quantity", with a mockup of the multiradial algorithm in which some parts are left as oracles; it is the prototype of the computer-science viewpoint of the present paper. [UHoTTa] and [UHoTTb] are overviews

of the dispute over identifications of isomorphic copies, in homotopy-type-theoretic vocabulary and by an educational model of indexed identifications; the present paper treats the same concern in the minimal form of factorization of maps, not in type-theoretic language. No theorem is imported from [UCS], [UHoTTa] or [UHoTTb].

What this paper does not claim

- It decides neither the truth nor the falsity of Corollary 3.12 or of the ABC conjecture.
- It does not claim an unconditional equivalence between the ABC conjecture itself and a property of forgetful functors; equivalence and reduction are stated only relative to the cited theory, uniform derivations, and extremal data.
- It makes no non-existence claim over the whole source. "Could not be located" is a report with the audited range and the reading made explicit.
- It does not present the elementary lemmas of the abstract skeleton as if they were deep theorems of IUT; the value lies in the instantiation and in the connection to the audit.

Reproducibility

The abstract core of §3—Definition 3.1, Propositions 3.2–3.4 and the examples of §2—is included in the accompanying repository[1] as the standalone Lean file `CompletenessFrameCore.lean`, which depends only on Mathlib and was checked within the standard axioms `propext`, `Classical.choice`, `Quot.sound` (the central restoration lemma uses none). The instantiation of the frame on value-part classes (`M3.relative.ActualValuePartCompletenessFrame`) and the sideways modules used in §6 are part of the author's larger Lean development, at commit `1e144663259ff07329f2158b3e2043d396fa4d54` of the branch `muse/valuepart-radial-20261110`; the frame file is reproduced in the repository for reference, and its selected build (341 modules) was verified within the standard three axioms. The source audit was carried out against checked Japanese HTML editions of [IUT1, IUT2, IUT3, IUT4, FrdI, EtTh, AbsTopIII] with mathematical notation structured in MathJax, with line-level references kept in the author's audit notes; the audit trail itself is not reproduced here. Independent review has not been received.

Disclosure

This paper was prepared with generative AI tools (large language models) under the author's direction, including the formalization, the audit, the writing and this English version, which the author prepared from his Japanese draft. Statements marked [EST] about the abstract frame are checked in Lean as described above. The correspondence between the frame and the source texts is an audit, not a Lean-checked assertion. The author takes full responsibility for the paper.

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